

A Compact Bias-Free ConvNeXt U-Net for Blind Gaussian Image Denoising

Frozen Gabor stem, Laplacian-pyramid skips, and end-to-end degree-1 homogeneity,
evaluated against the color-denoising leaderboard

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Abstract

We describe and evaluate BF-CONVUNEXT, a compact bias-free CONVNEXT U-Net for blind additive-white-Gaussian-noise (AWGN) image denoising. The network is an *integration* of four ingredients, none individually novel: a frozen depthwise **Gabor stem** (an oriented band-pass front end that contributes zero trainable parameters), a **Laplacian-pyramid** encoder that routes the full-resolution high-frequency residual into the skip connection (with three extra CONVNEXT blocks per skip band), a **CONVNEXT-V1 U-Net** body, and **end-to-end bias-freedom** — every convolution has no additive bias, the head is linear, activations are LeakyReLU, and normalization is a variance-only batch norm — so that the whole 3.39M-parameter network is exactly degree-1 homogeneous, $D(\alpha y) = \alpha D(y)$, at inference. Homogeneity is the structural property that licenses the Miyasawa/Tweedie reading of the denoiser residual as a (scaled) score, and, following Mohan et al., underwrites blind generalization across noise levels from a single model. We train one blind model with a linear noise- σ curriculum spanning $\sigma_{255} \approx 6$ –127 and evaluate it, unchanged, against the published color-denoising leaderboard on all four standard sets (CBSD68, Kodak24, McMaster, Urban100) at $\sigma \in \{15, 25, 50\}$ under the full-image protocol. The single blind model **beats DnCNN and FFDNet on every set at every noise level** (average +0.7 dB over DnCNN; FFDNet is non-blind yet still trails), and sits a small, consistent margin behind the heavyweight CNN/transformer state of the art — ~ 0.4 dB on CBSD68/Kodak24, ~ 0.7 dB on McMaster, ~ 1.0 dB on Urban100 — at roughly one tenth of their parameters. We are deliberately honest about what bias-freedom does *not* buy: the homogeneity is inference-only and checkpoint-specific, and the learned residual field is measurably non-conservative, so no global prior/energy exists and plug-and-play/RED convergence or calibrated-uncertainty guarantees do not transfer. We report PSNR and SSIM (SSIM for our own model, since the leaderboard we compare against is PSNR-only), on a single checkpoint, without ablations; those are stated as future work rather than glossed over.

1 Introduction

Blind Gaussian image denoising — recovering a clean image x from $y = x + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, \sigma^2 I)$, with σ *unknown at test time* — is both a canonical low-level vision task and a workhorse module inside modern inverse-problem solvers. Two facts frame this paper. First, the field’s accuracy is dominated by large transformer/CNN hybrids (DRUNet, SwinIR, Restormer, SCUNet) with tens of millions of parameters, often trained per-noise-level. Second, a much older and simpler observation — that removing *all* additive constants from a denoising CNN makes it exactly scale-equivariant and hence able to generalize blindly across noise levels a biased network cannot handle [Mohan et al., 2020] — remains underexploited in a modern architecture.

This paper takes the second fact seriously and asks a narrow, empirical question: *how far does a deliberately compact, strictly bias-free, modern-conv-block U-Net get on the standard color-denoising leaderboard, as a single blind model?* Our answer is BF-CONVUNEXT, a 3.39M-parameter bias-free CONVNEXT U-Net that composes four ideas we did not invent:

- a **frozen depthwise Gabor stem** [Gabor, 1946] — a bank of oriented, frequency-tuned band-pass filters fixed at initialization (zero trainable parameters), followed by a single learned 1×1 mixing projection;

- a **Laplacian-pyramid encoder** [Burt & Adelson, 1983] that, at each downsampling junction, sends the full-resolution high-frequency residual band to the skip connection and passes the low-frequency band downward, with three extra CONVNEXT blocks applied to each high-frequency skip band;
- a **CONVNEXT-V1 U-Net** [Liu et al., 2022, Ronneberger et al., 2015] body (depthwise 7×7 conv, inverted-bottleneck MLP, LayerScale);
- **end-to-end bias-freedom**: no additive bias in any layer, a linear head, LeakyReLU activations, and a variance-only batch normalization, chosen jointly so the network is globally degree-1 homogeneous at inference.

The scientific content is not that any of these pieces is new; it is (a) the specific, carefully bias-preserving way they are combined so that homogeneity survives end-to-end, and (b) an honest, apples-to-apples measurement of the result against the published leaderboard using the same test sets and full-image protocol. The headline is favorable: the compact blind model uniformly beats the classic sub-1M CNNs and lands about half a dB behind models an order of magnitude larger.

1.1 What this paper is (and is not)

This is an *integration-and-evaluation* paper in the register of an engineering report, not a novel-component paper. Specifically:

1. We contribute a precise description of BF-CONVUNEXT — the exact topology, the frozen Gabor stem, the Laplacian-pyramid skip decomposition, the CONVNEXT-V1 block, and the bias-free construction that makes the whole network degree-1 homogeneous — at a level of detail sufficient to reimplement it.
2. We contribute a fair benchmark: one blind checkpoint, evaluated unchanged on all four standard color sets at three noise levels under the full-image protocol, tabulated against six published baselines.
3. We are explicit about the theory’s *limits*: the Miyasawa/Tweedie residual-as-score reading is a motivation, and we show (Section 6) that the learned operator is non-conservative, so it is *not* the gradient of any global prior.

What this paper is *not*: it is not a claim that Gabor stems, Laplacian skips, CONVNEXT blocks, or bias-free CNNs are new; it is not an ablation study (we do not isolate each component’s contribution here — see Section 6); its comparison against the leaderboard is in **PSNR** (the collated source is PSNR-only), though we do report SSIM for our own model (Table 2); and it makes **no claim** of a calibrated prior, an exact likelihood, or plug-and-play convergence guarantees. We would rather under-claim on a single honest checkpoint than over-claim.

2 Architecture

BF-CONVUNEXT is a two-level U-Net operating on images normalized to $[-0.5, +0.5]$. All parameter counts and layer shapes below are read from the actual trained checkpoint (`convunext_denoiser_base`, 3,393,936 parameters total: 3,378,540 trainable and 15,396 frozen). Figure 1 shows the data flow.

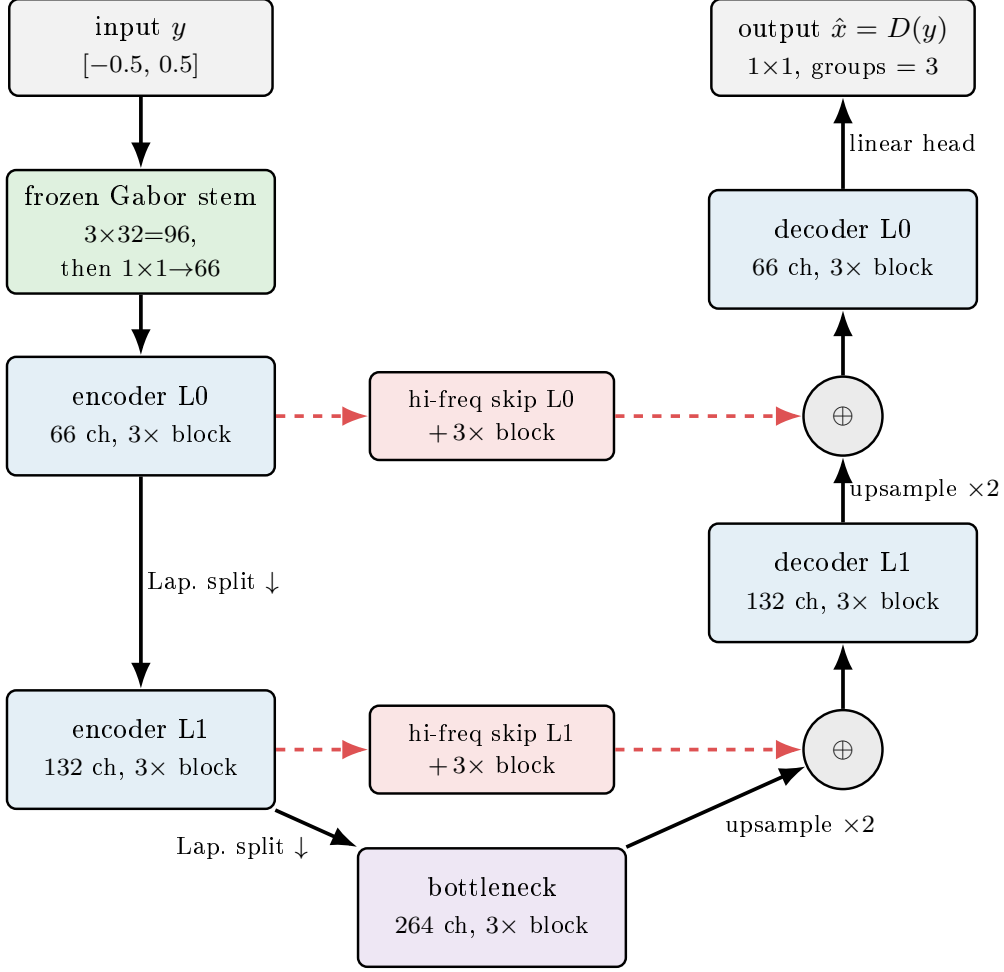


Figure 1: BF-CONVUNEXT data flow (channel widths shown). The encoder (left) descends; the decoder spine (right) ascends; the bottleneck sits at the base. A **frozen Gabor stem** feeds a two-level CONVNEXT-V1 U-Net. Downsampling is a **Laplacian-pyramid split**: the low-frequency band descends (solid arrows) while the full-resolution high-frequency residual becomes the skip (dashed), passing through **three extra bias-free CONVNEXT blocks**. At each decoder level the skip and the upsampled feature are first combined at an explicit parameter-free, channel-matched addition ($\oplus$) — a distinct step — and only then processed by the decoder block. The head is a bias-free grouped 1×1 convolution (3 groups). Every operation is degree-1 homogeneous at inference, so $D(\alpha y) = \alpha D(y)$.

2.1 U-Net topology

The channel widths follow $c_i = \text{round}(66 \cdot 2^i)$ for level i , giving $66 \rightarrow 132$ in the encoder, 264 at the bottleneck, and $132 \rightarrow 66$ in the decoder. Each of the five scopes (two encoder levels, bottleneck, two decoder levels) contains three CONVNEXT-V1 blocks. Upsampling is bilinear ($\times 2$). Channel counts are matched by a *parameter-free MatchChannels* operator (zero-pad to widen, slice to narrow) rather than a learned 1×1 projection, and decoder skips are fused by addition, $\text{dec} \leftarrow \text{Add}(\text{skip}, \text{MatchChannels}(\text{up}))$, not by concatenation. Both choices keep the merge exactly linear and homogeneous.

2.2 Frozen Gabor stem

The first learnable operation a naive CNN applies to a noisy image is a small learned convolution; here it is replaced by a *fixed* oriented band-pass front end. A depthwise convolution with 32 filters per input channel and a 7×7 kernel is initialized with a Gabor bank — each output channel is one 2D Gabor wavelet

$g(u, v) = \exp\left(-\frac{u_\theta^2 + \gamma^2 v_\theta^2}{2s^2}\right) \cos\left(\frac{2\pi u_\theta}{\lambda} + \psi\right)$ at a swept orientation/frequency/phase — and marked non-trainable, so it contributes **zero** trainable parameters and never adapts. With three input channels this emits $3 \times 32 = 96$ feature maps, which a single *learned* bias-free 1×1 convolution mixes down to the 66 stem channels. The rationale is standard signal processing: an oriented multi-scale band-pass decomposition is a strong, noise-agnostic image representation that the network need not spend capacity (or risk overfitting noise) to relearn.

2.3 Laplacian-pyramid skips and high-frequency blocks

Instead of max-pooling, each encoder junction performs a Laplacian-pyramid split. Given the level input z , a fixed Gaussian blur (5×5) and stride-2 blur-pool produce the low band $\ell = \text{blurpool}(\text{gauss}(z))$; the high band is the full-resolution residual $h = z - \text{up}(\ell)$. The split is lossless ($\text{up}(\ell) + h = z$) and channel-preserving. The low band ℓ descends to the next level; the high band h becomes the skip connection — but only after passing through **three** additional bias-free CONVNEXT-V1 blocks (**skip_highfreq**), applied at both encoder levels. This is a wavelet-shrinkage-style inductive bias: because Gaussian noise is spectrally flat while natural-image structure concentrates at low frequencies, forcing the decoder to reconstruct from an explicit low-band descent plus a *processed* high-band residual denies it the trivial full-resolution identity-copy shortcut and matches the signal/noise spectral structure of the task.

2.4 The CONVNEXT-V1 block

Each block is a residual branch with call order

$$x \mapsto x + \gamma \odot \left[W_\downarrow \phi \left(W_\uparrow \text{N}(\text{DW}_{7 \times 7} x) \right) \right],$$

where $\text{DW}_{7 \times 7}$ is a depthwise 7×7 convolution, N is the normalization (Section 2.5), $W_\uparrow$ is a 1×1 expansion to $4 \times$ width, ϕ is LeakyReLU with negative slope 0.1 (followed by dropout 0.1 at training time), $W_\downarrow$ is a 1×1 reduction, and γ is a per-channel LayerScale multiplier (initialized 10^{-4} , floored at 10^{-6}). *All* convolutions are bias-free. We use the V1 block rather than V2: the V2 variant inserts a Global Response Normalization whose learnable additive β is bias-like and would break strict bias-freedom.

2.5 Bias-free construction and degree-1 homogeneity

The defining property of BF-CONVUNEXT is that the entire map D is positively homogeneous of degree one at inference:

$$D(\alpha y) = \alpha D(y) \quad \text{for all } \alpha > 0. \quad (1)$$

This holds because every constituent operation is linear-through-the-origin or degree-1 homogeneous, and no operation injects an additive constant:

- every convolution (Gabor stem, 1×1 mix, all block convolutions, the head) has **use_bias=False**;
- the head activation is linear ($f(x) = x$);
- LeakyReLU is positively homogeneous, $\text{LReLU}(\alpha x) = \alpha \text{LReLU}(x)$ for $\alpha > 0$ (this is why LeakyReLU is used rather than GELU, which is not homogeneous);
- normalization is a *variance-only* batch norm, **BIASFREEBATCHNORM**: it has no running mean and no β offset, only a fixed non-trainable running variance, so at inference $\text{N}(x) = \gamma x / \sqrt{\text{Var} + \epsilon}$ is exactly linear in x ;
- the Laplacian split, **MatchChannels**, bilinear upsampling, and additive skip merge are all linear.

The choice of **BIASFREEBATCHNORM** over the CONVNEXT default LayerNorm is essential and not cosmetic: LayerNorm divides by a per-sample statistic that itself scales with the input, making it *scale-invariant* (degree-0), which would destroy Eq. (1) for the whole network. We flag one scoping caveat that most treatments omit: homogeneity is an *inference-time* property. During training **BIASFREEBATCHNORM** uses the per-batch variance, which makes it degree-0; Eq. (1) holds once the running statistics are frozen.

3 Why bias-free: Miyasawa/Tweedie, and its limits

Bias-freedom is not an aesthetic choice; it is what connects the trained denoiser to a score, and what lets one blind model span a wide noise range. We state the motivation and then, in the same breath, its boundaries.

Miyasawa/Tweedie. For $y = x + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \sigma^2 I)$ and marginal density $p_\sigma(y) = \int \mathcal{N}(y; x, \sigma^2 I) p(x) dx$, the minimum-MSE estimator obeys the Miyasawa/Tweedie identity

$$\hat{x}(y) = \mathbb{E}[x | y] = y + \sigma^2 \nabla_y \log p_\sigma(y). \quad (2)$$

Hence the denoiser *residual* $f(y) = D(y) - y$ is, for an MMSE-optimal D , exactly σ^2 times the score $\nabla_y \log p_\sigma(y)$ of the noise-blurred density. Training with a plain mean-squared-error objective is therefore not an arbitrary loss: it drives D toward the MMSE estimator whose residual is (a scaled) score. This is the reason BF-CONVUNEXT is trained with MSE and read as an implicit-prior score model.

Homogeneity buys blind generalization. Mohan et al. [2020] observe that a bias-free denoiser is degree-1 homogeneous (Eq. (1)) and therefore behaves as a locally linear, input-adaptive filter $D(y) \approx J(y) y$ whose Jacobian $J(y) = \partial D / \partial y$ is degree-0 (scale-invariant). Because J depends only on the *direction* of y , not its magnitude, a network trained over one range of noise levels extrapolates gracefully to unseen levels — whereas a biased network, whose additive terms fix an absolute scale, collapses off its training range. This is precisely the property that makes a single blind model viable, and the empirical target of our noise curriculum (Section 4).

What homogeneity does *not* buy. We are careful here, because it is easy to over-read Eq. (2). Two boundaries hold and are stated as first-class caveats:

1. **Homogeneity is checkpoint-specific, not an architectural guarantee.** A network is exactly homogeneous only if it is *truly* bias-free in every operation. For this checkpoint the relative homogeneity error $\|D(\alpha y) - \alpha D(y)\| / \|\alpha D(y)\|$ is at the level of floating-point round-off across $\alpha \in \{\frac{1}{4}, \frac{1}{2}, 2, 4, 8\}$; but sibling checkpoints of the same family have shown deviations up to ~ 0.14 when a single non-homogeneous operation slips in. Eq. (1) must be *verified* per checkpoint, not assumed.
2. **The learned residual is not a conservative field.** Eq. (2) would make $f(y)$ the gradient of the scalar $\sigma^2 \log p_\sigma(y)$ only if the Jacobian $J(y)$ were symmetric. It is not: on trained checkpoints of this family the measured Jacobian asymmetry $\|J - J^\top\| / \|J\|$ is $\mathcal{O}(0.5)$ — roughly four orders of magnitude above a symmetric-blur control — so **no global energy/log-density function exists** whose gradient is the residual. Consequently, plug-and-play and Regularization-by-Denoising convergence guarantees (which assume a conservative, symmetric-Jacobian denoiser) do not transfer, and the residual should be read as a useful *local* score estimate, not a calibrated global prior.

We return to the practical consequences (including uncertainty calibration) in Section 6.

4 Experimental protocol

Data and noise curriculum. We train on 256×256 patches drawn from COCO (train2017) and DIV2K (train), sourced with a per-directory weighting so the $\sim 118\text{K}$ COCO images do not drown DIV2K’s ~ 800 . Patches are normalized to $[-0.5, +0.5]$ (peak-to-peak range 1.0) and corrupted with additive Gaussian noise whose per-patch standard deviation is drawn uniformly from $[0, \sigma_{\max}]$, where the upper bound $\sigma_{\max}$ is widened linearly over training by a curriculum callback from 0.025 to 0.5 across 100 epochs. Starting narrow and widening stabilizes early optimization; bias-freedom then supplies the cross-level generalization.

The σ convention (stated precisely). The model operates entirely in $[-0.5, +0.5]$, so its noise level $\sigma_{\text{norm}} \in [0, 0.5]$. The literature reports noise on the $[0, 255]$ pixel scale; the two are related by $\sigma_{255} = \sigma_{\text{norm}} \cdot 255$. The curriculum’s $\sigma_{\max} : 0.025 \rightarrow 0.5$ therefore spans $\sigma_{255} \approx 6.4 \rightarrow 127.5$ — it covers and

exceeds the classic 15/25/50 benchmark regimes as a single blind model. All PSNR values are computed with `max_val = 1.0` (matched to the peak-to-peak range); because PSNR is scale-invariant under a matched range, these dB numbers are directly comparable to published `max_val = 255` results, with no further conversion.

Optimization. Loss is plain MSE (the Miyasawa-optimal objective, Section 3). We use AdamW (learning rate 10^{-3} , cosine decay, 10 warmup epochs, decoupled weight decay 4×10^{-3}), gradient-norm clipping at 1.0, batch size 4, 100 epochs, full `fp32` (mixed precision was measured slower on this convolution-heavy U-Net because the bilinear-upsample gradient forces XLA off).

Evaluation. We report two evaluations of the *single* trained checkpoint. (i) A blind PSNR-vs-noise sweep on DIV2K-validation (100 patches at 256px). (ii) The standard leaderboard comparison on all four color sets (CBSD68, Kodak24, McMaster, Urban100) at $\sigma \in \{15, 25, 50\}$ using the **full-image protocol** (whole images, reflect-padded to a multiple of 16), the same protocol the published numbers use. We additionally report a post-hoc per-pixel split-conformal coverage study. Two evaluation samples differ (100 vs. 32 patches for the sweep vs. the coverage study), so their PSNR columns are independent measurements, not cross-checks of one another.

5 Results

5.1 Blind denoising across noise levels

Figure 2 shows the blind PSNR-vs-noise sweep on DIV2K-validation. A single model, given no noise-level input, denoises across the whole curriculum range, from 40.7 dB at $\sigma_{255}=5$ down to 29.1 dB at $\sigma_{255}=65$, improving the noisy input by +6.4 to +15.8 dB.

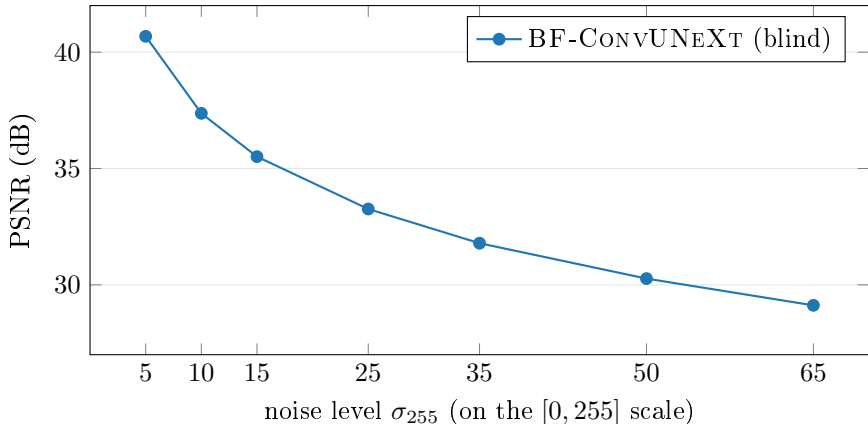


Figure 2: Blind PSNR vs. noise on DIV2K-validation (100 patches, 256px). One model spans $\sigma_{255} \approx 5$ –65 with no noise-level input; 95% bootstrap CIs (not shown) are ± 0.4 –0.8 dB.

5.2 Comparison against the color-denoising leaderboard

Table 1 places BF-CONVUNEXT against six published baselines on all four standard color sets at $\sigma \in \{15, 25, 50\}$, under the identical full-image protocol and PSNR convention. The pattern is uniform: the single 3.39M blind model **beats DnCNN and FFDNet on every set at every noise level**, and trails the heavyweight state of the art by a small, consistent margin that widens on Urban100.

Table 1: Color AWGN denoising, PSNR (dB), full-image protocol. BF-CONVUNEXT is a single 3.39M-parameter blind model (no noise-level input). Baseline numbers are the published values collated by Zhang et al. [2022]. **Bold** marks where BF-CONVUNEXT beats a given baseline. Approximate parameter counts are listed in the header; DnCNN/FFDNet are < 1M, the four rightmost are 12–33M.

Set	σ_{255}	BF-CONVUNEXT (3.4M)	DnCNN (0.6M)	FFDNet (0.5M)	DRUNet (32M)	SwinIR (12M)	Restormer (26M)	SCUNet (18M)
CBSD68	15	33.95	33.90	33.87	34.30	34.42	34.40	34.40
CBSD68	25	31.35	31.24	31.21	31.69	31.78	31.79	31.79
CBSD68	50	28.18	27.95	27.96	28.51	28.56	28.60	28.61
Kodak24	15	34.97	34.60	34.63	35.31	35.34	35.47	35.34
Kodak24	25	32.56	32.14	32.13	32.89	32.89	33.04	32.92
Kodak24	50	29.51	28.95	28.98	29.86	29.79	30.01	29.87
McMaster	15	34.79	33.45	34.66	35.40	35.61	35.61	35.60
McMaster	25	32.61	31.52	32.35	33.14	33.20	33.34	33.34
McMaster	50	29.61	28.62	29.18	30.08	30.22	30.30	30.29
Urban100	15	34.20	32.98	33.83	34.81	35.13	35.13	35.18
Urban100	25	31.92	30.81	31.40	32.60	32.90	32.96	33.03
Urban100	50	28.81	27.59	28.05	29.61	29.82	30.02	30.14
avg $\sigma=15$		34.48	33.73	34.25	34.96	35.13	35.15	35.13
avg $\sigma=25$		32.11	31.43	31.77	32.58	32.69	32.78	32.77
avg $\sigma=50$		29.03	28.28	28.54	29.52	29.60	29.73	29.73

Averaged over the four sets, BF-CONVUNEXT is +0.75/+0.68/+0.75 dB above DnCNN at $\sigma = 15/25/50$ and +0.23/+0.34/+0.49 dB above FFDNet (which, unlike our model, is *non-blind* and receives a noise-level map). It is −0.48/−0.47/−0.49 dB relative to DRUNet and −0.67/−0.67/−0.70 dB relative to Restormer, the leaderboard top. The gap is smallest on CBSD68/Kodak24 (~ 0.4 dB), larger on McMaster (~ 0.7 dB), and largest on Urban100 (~ 1.0 dB) — Urban100’s repetitive, self-similar structures are exactly where the transformers’ non-local attention (SwinIR, Restormer) and DRUNet’s large receptive field have the most to exploit and a compact local conv U-Net the least.

5.3 Structural similarity (SSIM)

Table 2 reports SSIM alongside PSNR for BF-CONVUNEXT on the same four sets and full-image protocol (same checkpoint as Table 1), computed on the $[0, 1]$ representation with a data range of 1.0. SSIM tracks PSNR’s ordering across noise — highest at $\sigma_{255}=15$ (average 0.930) and degrading gracefully to average 0.827 at $\sigma_{255}=50$. One cross-metric wrinkle is worth noting: **Urban100 has the *highest* SSIM of the four sets at every noise level** (0.944/0.918/0.866) despite its middling PSNR, because its regular, high-contrast man-made structures — the same self-similarity that costs PSNR against non-local models — yield strong local structural agreement once denoised. We report SSIM for BF-CONVUNEXT only: our comparison source [Zhang et al., 2022] tabulates PSNR alone, and SSIM values assembled from mixed sources use inconsistent windowing/border/color conventions, so a baseline SSIM column would be an unreliable comparison rather than an honest one.

Table 2: BF-CONVUNEXT PSNR (dB) and SSIM on the four color benchmark sets, full-image protocol, single blind checkpoint (identical protocol and model as Table 1; the PSNR columns reproduce it exactly). SSIM on $[0, 1]$, data range 1.0. Baseline SSIM is not tabulated — see Section 5.3.

Set	$\sigma_{255} = 15$		$\sigma_{255} = 25$		$\sigma_{255} = 50$	
	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
CBSD68	33.95	0.9311	31.35	0.8873	28.18	0.8010
Kodak24	34.97	0.9263	32.56	0.8870	29.51	0.8112
McMaster	34.79	0.9181	32.61	0.8879	29.61	0.8287
Urban100	34.20	0.9437	31.92	0.9182	28.81	0.8661
<i>average</i>	34.48	0.9298	32.11	0.8951	29.03	0.8268

5.4 Accuracy for the parameter budget

The point of Table 1 is not that BF-CONVUNEXT wins outright — it does not — but where it sits on the accuracy/size trade-off. Figure 3 plots the four-set average PSNR at $\sigma = 25$ against approximate parameter count. BF-CONVUNEXT lands between the two size classes: comfortably above the sub-1M classic CNNs it beats, and within half a dB of models 8–10 $\times$ larger.

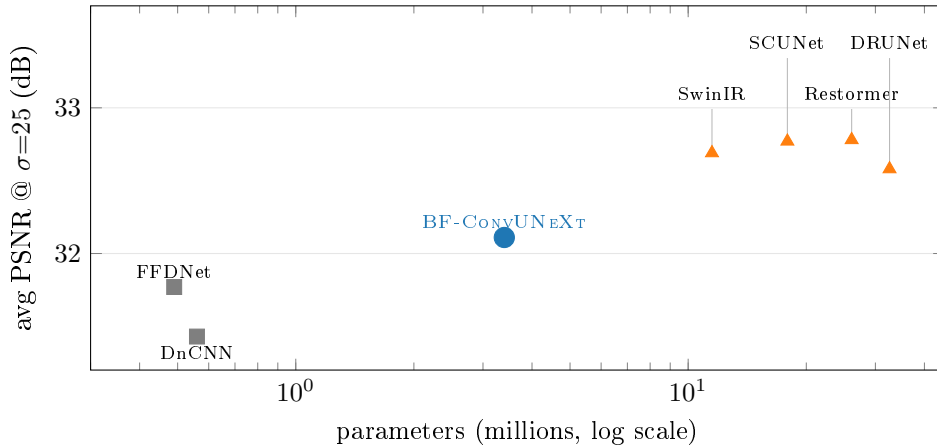


Figure 3: Four-set average PSNR at $\sigma=25$ vs. (approximate) parameter count. BF-CONVUNEXT (blue) sits on the efficient side of the trade-off: above the sub-1M CNNs and within ~ 0.5 – 0.7 dB of the 12–33M state of the art at $\sim 1/8$ – $1/10$ of their size. Parameter counts are approximate and collated from the respective papers.

5.5 Per-pixel uncertainty: honest coverage

Because the residual is a local score estimate, a natural question is whether a simple post-hoc interval calibrates. We ran split-conformal per-pixel intervals (per- σ Mondrian, target coverage 0.90, $\alpha = 0.1$, 32 calibration images) on the checkpoint; Table 3 reports empirical coverage, mean interval width, and the matched test-split PSNR. Coverage is systematically ~ 0.877 , about 2.3 points below the 0.90 target.

Table 3: Post-hoc split-conformal per-pixel coverage (target 0.90). Coverage is systematically below target; see text for why this is expected, not a bug.

σ_{norm}	empirical coverage	mean width	test PSNR (dB)
0.05	0.877	0.053	35.08
0.10	0.879	0.076	31.73
0.15	0.878	0.093	29.80
0.20	0.878	0.107	28.45
0.25	0.877	0.119	27.38

We resist the temptation to call this finite-sample noise. With 32 images the calibration pool is $\sim 6.3\text{M}$ pixels, so the conformal quantile is estimated essentially exactly; a well-specified split conformal would sit at 0.90. The systematic ~ 0.877 instead reflects a genuine exchangeability gap: a *single* per- σ quantile pools pixels across images whose per-image residual scales are heterogeneous, so pixel-level exchangeability requires image-level exchangeability, which does not fully hold on held-out images. Image-conditional or variance-scaled conformal would tighten it; that is out of scope here. We report it because a paper that shows the coverage table but hides the undercoverage would be dishonest.

6 Discussion: limitations and what does not work

Where it wins, and where it does not. BF-CONVUNEXT is a strong *compact, blind* denoiser: it beats the classic sub-1M CNNs everywhere and closes most of the gap to models an order of magnitude larger, from a single checkpoint with no noise-level input. It does *not* beat the heavyweight state of the art, and the gap is structurally largest on Urban100, where non-local self-similarity rewards attention and large receptive fields that a two-level local conv U-Net lacks. We do not claim otherwise.

The theory’s limits are real, not rhetorical. As shown in Section 3, the learned residual field is measurably non-conservative, so BF-CONVUNEXT does *not* define a global image prior, an exact likelihood, or a calibrated posterior. Uses that require a conservative denoiser — provable plug-and-play/RED convergence, energy-based sampling with a fixed target, calibrated Bayesian uncertainty — are outside what this model licenses, and the undercoverage in Section 5.5 is a concrete downstream symptom. The homogeneity that does hold is inference-only and must be re-verified per checkpoint rather than assumed from the architecture.

Measurement limits. We report PSNR and SSIM for BF-CONVUNEXT (Tables 1 and 2), but the *comparison* against baselines is **PSNR-only**, because our collated source [Zhang et al., 2022] reports PSNR and cross-source SSIM is protocol-inconsistent; a perceptual metric (e.g. LPIPS) is not measured at all. All results are from a **single checkpoint** — there is no multi-seed variance estimate — and we present **no ablations**: the individual contributions of the frozen Gabor stem, the Laplacian high-frequency skips, the **high_freq_blocks**, the bias-free normalization, and the noise curriculum are *not* isolated here. CBSD68 and McMaster are the two standard color sets we could evaluate on locally alongside Kodak24 and Urban100; grayscale Set12/BSD68 would require a one-channel model. Each of these is deliberate scope, and each is the obvious next experiment.

Future work. The natural follow-ups are (i) a controlled ablation isolating each component’s dB contribution with multi-seed variance; (ii) a perceptual metric (e.g. LPIPS) alongside the PSNR/SSIM already reported, and baseline SSIM under a single re-run protocol so the comparison can extend beyond PSNR; (iii) larger-capacity and deeper variants to probe how much of the SOTA gap is capacity vs. architecture; and (iv) a study of whether a symmetrized or explicitly conservative variant recovers the plug-and-play guarantees the current model forgoes, and at what accuracy cost.

7 Related work

Bias-free denoising. Mohan et al. [2020] introduced bias-free CNNs for blind denoising and established the degree-1-homogeneity argument we build on. The residual-as-score reading rests on the classical Miyasawa/Tweedie identity [Miyasawa, 1961, Robbins, 1956], which also underpins the implicit-prior sampling line of Kadkhodaie & Simoncelli [2021]. BF-CONVUNEXT applies the bias-free discipline end-to-end to a modern block and front end.

CNN and transformer denoisers. DnCNN [Zhang et al., 2017] established the residual-CNN denoiser; FFDNet [Zhang et al., 2018] added a noise-level map for non-blind flexibility; DRUNet [Zhang et al., 2021] coupled a U-Net denoiser with plug-and-play restoration; SwinIR [Liang et al., 2021], Restormer [Zamir et al., 2022], and SCUNet [Zhang et al., 2022] brought (windowed/channel) attention and are the current leaderboard leaders. These are our baselines; all except our model and DnCNN’s blind variant either use a noise map or are larger by 8–60 $\times$.

Building blocks. The block is a CONVNEXT [Liu et al., 2022] in a U-Net [Ronneberger et al., 2015]. The multi-scale skip decomposition is a Laplacian pyramid [Burt & Adelson, 1983], and the frozen front end is a Gabor filter bank [Gabor, 1946]. BF-CONVUNEXT contributes their bias-preserving integration and its measurement, not the pieces.

8 Reproducibility

The evaluated checkpoint is `convunext_denoiser_base` (3,393,936 parameters; depth 2; widths 66/132/264; three CONVNEXT-V1 blocks per scope plus three per high-frequency skip band; frozen 32-filter, 7 $\times$ 7 Gabor stem; BIASFREEBATCHNORM; LeakyReLU(0.1); linear grouped head). It was trained with the recipe of Section 4. The full-image benchmark uses whole test images reflect-padded to a multiple of 16, PSNR at `max_val` = 1.0 on $[-0.5, 0.5]$, on the standard CBSD68 / Kodak24 / McMaster / Urban100 sets. All numbers in Table 1, Table 3, and Figure 2 are measured from that single checkpoint; baseline numbers are the published values as collated by Zhang et al. [2022].

9 Conclusion

We described BF-CONVUNEXT, a compact bias-free CONVNEXT U-Net that integrates a frozen Gabor stem, Laplacian-pyramid high-frequency skips, and end-to-end degree-1 homogeneity, and we measured it honestly against the color-denoising leaderboard. As a single 3.4M-parameter blind model it beats the classic sub-1M CNNs on all four standard sets at all three noise levels and comes within about half a dB of state-of-the-art models an order of magnitude larger, with the residual gap concentrated on self-similar imagery. We were equally explicit about the boundaries: the bias-free homogeneity is inference-only and checkpoint-specific, the learned operator is non-conservative and thus not a global prior, and we report a single checkpoint on PSNR without ablations. The contribution is a clean, reproducible integration and a fair benchmark, not a novelty claim — and, we hope, a useful data point on how much of modern denoising accuracy is reachable from a small, strictly bias-free model.

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